

Unit 2 Classwork — Exponents & Scientific Notation

Exponent Rules · Power/Zero/Negative Exponents · Scientific Notation

A. Exponent Rules — Product & Quotient

Problem 1

Simplify: $x^4 \cdot x^3$

answer _____

Problem 2

Simplify: $y^8 \div y^3$

answer _____

Problem 3

Simplify: $(2x^3)(3x^2)$

answer _____

Problem 4

Simplify: $15a^6 \div 3a^2$

answer _____

Problem 5

Simplify: $a^2 \cdot a^5 \cdot a$

answer _____

B. Power, Zero & Negative Exponents

Problem 1

Simplify: $(x^3)^4$

answer _____

Problem 2

Simplify: $(2x^2)^3$

answer _____

Problem 3

Simplify: 7^0

answer _____

Problem 4

Simplify: $(5x^4)^0$

answer

Problem 5

Simplify: 3^{-2} (write as a fraction)

answer

Problem 6

Simplify: x^{-4} (write as a fraction)

answer

Problem 7

Evaluate: -4^2 vs $(-4)^2$ — what is $(-4)^2$?

answer

C. Scientific Notation — Conversions**Problem 1**

Write 32,000 in scientific notation.

answer

Problem 2

Write 0.0091 in scientific notation.

answer

Problem 3

Write 5.6×10^6 in standard form.

answer

Problem 4

Write 2.3×10^{-4} in standard form.

answer

Problem 5

Write 470,000,000 in scientific notation.

answer

D. Scientific Notation — Operations

Problem 1

Multiply: $(2 \times 10^3) \times (3 \times 10^5)$

answer

Problem 2

Divide: $(8 \times 10^7) \div (2 \times 10^3)$

answer

Problem 3

Multiply: $(6 \times 10^4) \times (5 \times 10^2)$ — adjust if needed!

answer
